PIERCE: **P**erception-**I**ntegrated **E**xecution in **R**obotic **C**ulinary **E**nvironments

EECS 467: Autonomous Robotics, Winter 2026

Xiao, Qixin
University of Michigan
qxiaocs@umich.edu

Xi, Avery
University of Michigan
axi@umich.edu

Mann, Elaina
University of Michigan
elainamn@umich.edu

Sundher, Medhya
University of Michigan
msundher@umich.edu

***Abstract*—Bimanual manipulation of soft, contact-rich objects remains challenging for low-cost robotic systems, especially in culinary tasks such as fruit skewering, where the robot must coordinate precise grasping, alignment, and insertion under visual and physical uncertainty. We present PIERCE, Perception-Integrated Execution in Robotic Culinary Environments, a low-cost dual-arm robotic system for autonomous fruit skewering. The system uses a 360° rotating base, custom skewer-holding hardware, multi-view RGB observations, and robot proprioception to perform bimanual manipulation without relying on expensive tactile sensors. PIERCE learns manipulation behaviors from expert demonstrations collected through a leader-follower teleoperation pipeline. A pretrained Vision-Language-Action policy is fine-tuned on these demonstrations and deployed in a closed-loop perception-action system with real-time action chunking. We evaluate the system through staged subtask testing and final autonomous operation, including object grasping, alignment, skewering, dropping, and color-based sorting. The results show that modern visuo-motor imitation learning can be adapted to a low-cost bimanual platform for precise contact-rich culinary manipulation, while also revealing key limitations in generalization, memory, and robustness under distribution shift.**

***Index Terms*—Bimanual manipulation, contact-rich manipulation, imitation learning, Vision-Language-Action models, teleoperation, culinary robotics, multi-view perception.**

I. Introduction

Robotic manipulation in everyday environments remains difficult because real-world objects are rarely rigid, uniform, or arranged in perfectly controlled settings. Culinary manipulation is a representative example of this difficulty. Food items vary in size, shape, surface texture, softness, and friction, and they often require physical contact that is hard to model precisely. A task such as fruit skewering appears simple for humans, but it requires a robot to locate small objects, grasp them without dropping or crushing them, align them with a narrow skewer, and apply enough force to insert them successfully.

This project focuses on autonomous fruit skewering with a low-cost bimanual robot. The task is contact-rich and coordination-heavy: one arm must stabilize the skewer while the other arm picks up the fruit, aligns it with the skewer tip, and performs the insertion motion. Small pose errors can cause the fruit to miss the skewer, slip out of the gripper, or deform during contact. These challenges are especially significant because our system does not use expensive tactile sensors. Instead, it relies on multi-view visual observations, robot proprioception, and learned visuo-motor control.

We propose PIERCE, short for Perception-Integrated Execution in Robotic Culinary Environments. PIERCE is a dual-arm robotic system mounted on a 360° rotating base. The system uses multiple RGB camera views to observe the workspace and a custom skewer gripper to stabilize the target during insertion. The two arms are assigned complementary roles: the left arm holds and stabilizes the skewer, while the right arm performs object grasping, alignment, and insertion. This bimanual role decomposition makes the task more tractable than attempting to complete the entire manipulation sequence with a single arm.

The core control policy is based on a pretrained Vision-Language-Action model from the LeRobot framework, specifically SmolVLA [1], [2]. We fine-tune this policy using imitation learning from expert demonstrations collected through a leader-follower teleoperation setup. During data collection, a human operator controls leader arms while the robot follower arms reproduce the demonstrated motions. This allows us to collect coordinated bimanual demonstrations without manually programming each motion primitive. To make training cleaner, safer, and more repeatable, foam balls are used as synthetic proxies for grapes during data collection, while final testing is performed on real fruit.

The main contributions of this project are as follows:

- We design and build a low-cost bimanual robotic system for contact-rich fruit skewering.
- We develop a hardware architecture with complementary arm roles, a custom skewer gripper, multi-view cameras, and a 360° rotating base.
- We collect teleoperated bimanual demonstrations and fine-tune a pretrained Vision-Language-Action policy for culinary manipulation.
- We deploy the learned policy in a real-time closed-loop system with action chunking, a FastAPI runtime server, and a web dashboard for monitoring and control.
- We evaluate the system through staged subtask testing and final autonomous operation, identifying both successful behaviors and remaining limitations.

II. Related Work

A. Contact-Rich and Culinary Manipulation

Culinary robotics has become an important testbed for manipulation because food preparation tasks involve deformable objects, uncertain contact, and high variability in object geometry. Unlike rigid industrial parts, food items can slip, compress, deform, or break during manipulation. Prior work in robotic food handling has explored how specialized manipulation strategies can improve robustness in these settings. For example, the CARBS framework uses a secondary arm as a physical constraint during food manipulation, showing that an additional limb can simplify the control problem by stabilizing the object or environment [3]. Other work on robotic stir-frying demonstrates that coordinated arm motions can be useful for cooking tasks that require repeated contact and temporal coordination [4].

PIERCE follows the same general motivation but targets a different culinary skill: fruit skewering. Compared with scooping or stirring, skewering requires accurate axial alignment between the fruit and the skewer tip, followed by a controlled insertion motion through a soft object. This makes the task particularly sensitive to small visual and positional errors. Our system addresses this challenge by combining bimanual role decomposition, multi-view visual feedback, and learned visuomotor control.

B. Bimanual Manipulation and Teleoperation

Bimanual manipulation is useful when a task requires one hand to stabilize an object while the other hand performs a precise action. This structure is common in human manipulation and is especially important for contact-rich tasks. In PIERCE, the left arm stabilizes the skewer, while the right arm grasps and inserts the fruit. This separation of roles reduces the burden on each individual arm and makes the insertion process more stable.

Collecting demonstrations for bimanual manipulation is difficult because the motions of the two arms must be temporally coordinated. Teleoperation provides a practical solution by allowing a human operator to directly demonstrate the desired behavior. Prior robot learning systems have shown that teleoperated demonstrations can provide high-quality supervision for imitation learning, especially when the target behavior is difficult to manually specify through analytical control rules [5]. We adopt a leader-follower teleoperation setup so that the operator can demonstrate smooth bimanual motions while the robot records synchronized observations and actions.

C. Vision-Based Imitation Learning and VLA Policies

Imitation learning is a common approach for training robots to perform complex manipulation skills from expert demonstrations. Instead of designing hand-written controllers for every stage of the task, imitation learning trains a policy to map observations directly to actions. This approach is attractive for food manipulation because contact dynamics and object deformation are difficult to model explicitly.

Recent Vision-Language-Action models extend this idea by conditioning robot actions on visual observations, proprioceptive states, and natural language task instructions. These models provide a flexible framework for learning multiple manipulation behaviors under a shared policy structure [6]–[8]. SmolVLA, implemented in the open-source LeRobot framework, provides a practical pretrained backbone for low-cost robot learning and supports action chunking for temporally consistent control [1], [2]. PIERCE builds on this direction by fine-tuning a pretrained VLA policy on our own teleoperated demonstrations and deploying it on a physical bimanual platform.

D. Position of PIERCE

PIERCE combines ideas from culinary manipulation, bimanual role decomposition, teleoperation, and VLA-based imitation learning. Its main distinction is the specific combination of low-cost hardware and precise contact-rich fruit skewering. Rather than relying on tactile sensing or manually scripted insertion policies, the system learns grasping, alignment, and skewering behavior from demonstrations. This makes PIERCE a practical exploration of how modern visuo-motor policies can be adapted to affordable robotic systems for real-world culinary manipulation.

III. System Overview

A. Task Definition

PIERCE, short for Perception-Integrated Execution in Robotic Culinary Environments, is a low-cost bimanual robotic system for autonomous fruit skewering. The task requires the robot to visually perceive food objects, grasp a selected item, align it with a skewer, insert it onto the skewer, and optionally drop or sort objects into target containers. In addition to the full end-to-end skewering task, the system is designed to support several isolated manipulation skills, including object detection, object pickup, skewering, object dropping, and color-based sorting.

This task is challenging because skewering is a contact-rich manipulation problem. The robot must coordinate two arms while interacting with soft and deformable objects, where small alignment errors can cause missed contact, object slipping, or fruit deformation. Since our goal is to build an affordable system, PIERCE avoids expensive tactile sensing and instead relies mainly on visual observations, robot proprioception, and learned visuo-motor control.

B. System Architecture

The overall system follows a sensing-planning-acting loop. In the sensing stage, RGB camera streams and robot proprioceptive states are used to construct the observation at each timestep. In the planning and learning stage, a vision-based imitation learning policy maps these observations, together with a natural language task instruction, into robot action commands. In the acting stage, the dual-arm robot executes the predicted actions through the left and right arms and updates the workspace state. The changed environment is

TABLE I
MAIN HARDWARE COMPONENTS USED IN PIERCE.

Component	Role in the System
Dual SO101 arms	Provide bimanual manipulation for stabilizing the skewer and grasping/inserting objects.
Parallel grippers	Enable grasping of fruit pieces or synthetic proxy objects.
Custom skewer gripper	Holds and stabilizes the skewer during insertion.
360° rotating base	Expands reachable workspace and improves approach flexibility.
Suction cup mount	Improves physical stability during manipulation.
RGB cameras	Provide visual observations for perception and policy inference.
Local GPU workstation	Runs real-time model inference and robot control.

then observed again, forming a closed-loop perception-action pipeline.

The two arms play complementary roles. The left arm mainly stabilizes the skewer and holder, while the right arm performs object grasping, alignment, and insertion. This division of labor reduces the difficulty of the contact-rich insertion process, because the skewer can remain fixed while the other arm performs the precise motion required to add the fruit.

C. Hardware Components and Materials

The hardware platform consists of two SO101 robotic arms mounted on a 360° rotating base. The robot uses parallel grippers for object manipulation and a custom skewer gripper for holding the skewer more reliably during insertion. A suction cup mount is used to improve base stability. The system also includes multiple RGB camera views, including a head camera and wrist-mounted camera views, to observe the workspace and reduce occlusion during close-range bimanual manipulation.

IV. METHODOLOGY

A. Mechanical Design

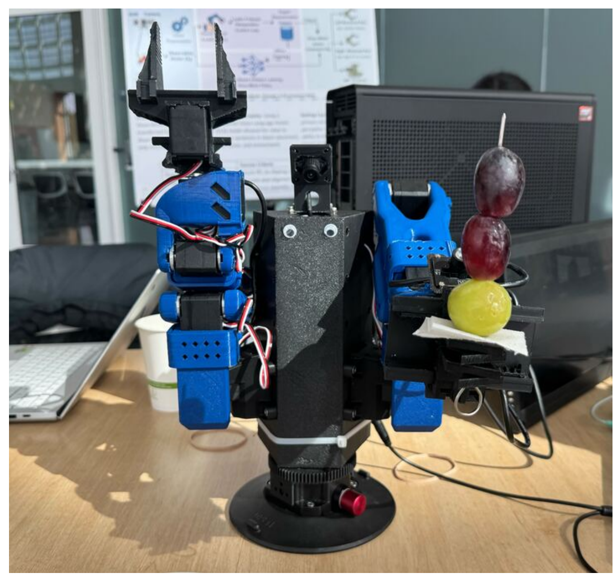
The mechanical design of PIERCE is centered on making contact-rich fruit skewering feasible with low-cost hardware. The system uses a dual-arm robot mounted on a 360° rotating base. This design increases workspace coverage and allows the robot to approach objects from more flexible angles. Since the goal of the project is to avoid expensive tactile sensors, the hardware is designed to rely primarily on visual feedback, robot proprioception, and learned visuo-motor control.

The two arms are assigned complementary roles. The left arm is responsible for stabilizing the skewer and maintaining a consistent target pose during insertion. The right arm is responsible for grasping the fruit or proxy object, aligning it with the skewer tip, and executing the insertion motion. This division of labor simplifies the task because the robot does not need one arm to both stabilize the target and perform the contact-rich manipulation. It also reduces unwanted motion of the skewer during piercing, which is important for soft and deformable objects such as grapes.

Several custom hardware choices were added to improve task reliability. A custom skewer gripper constrains the skewer pose and reduces slipping during insertion. A suction cup mount improves base stability during manipulation. The rotating base gives the robot additional flexibility when objects or cups are arranged around the workspace rather than directly in front of the arms. The perception system uses multiple RGB camera views, including a head camera and wrist-mounted camera views, to observe the workspace, the grippers, the target object, and the skewer during close-proximity bimanual interaction.

(a) Custom skewer gripper

(b) Rotating base



(c) Full PIERCE platform

Fig. 2. Mechanical components of the PIERCE platform. The custom skewer gripper stabilizes the skewer during insertion, the rotating base improves workspace flexibility, and the full platform integrates the bimanual arms, skewer holder, camera setup, and support structure.

B. Teleoperation Data Collection

Training data was collected using a leader-follower teleoperation pipeline. During data collection, a human operator controlled the leader arms, and the follower robot reproduced the demonstrated bimanual motions in real time. This setup allowed the operator to provide expert demonstrations for grasping, alignment, skewering, dropping, and sorting behaviors. Each demonstration was recorded as an observation-

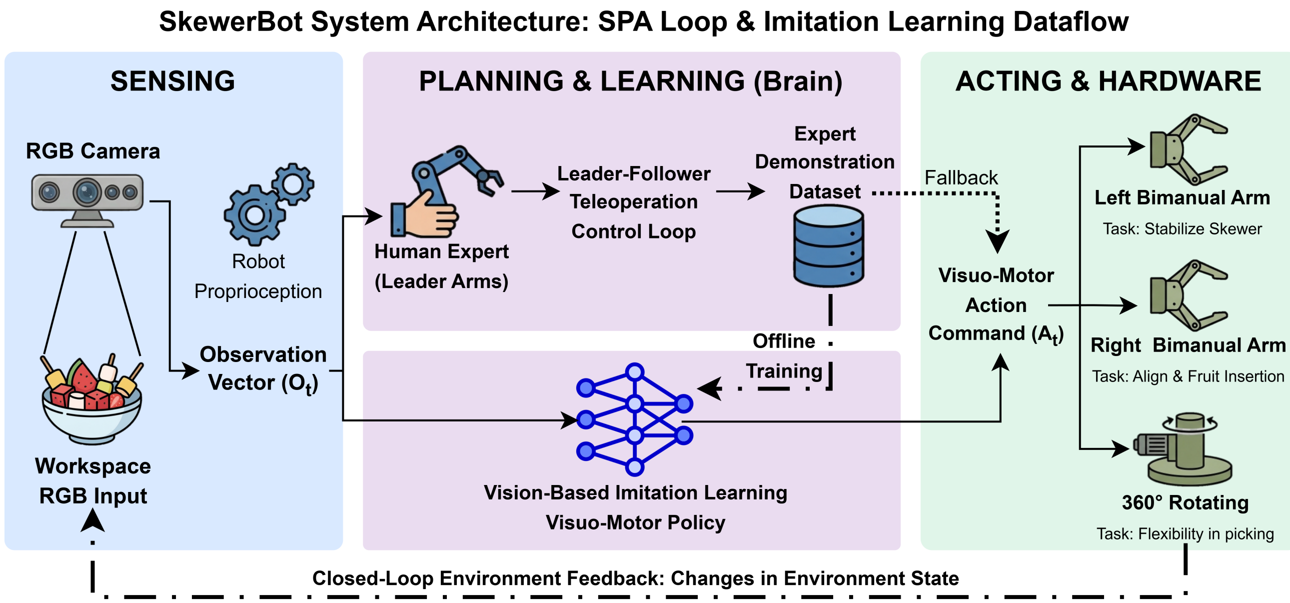


Fig. 1. PIERCE system architecture. The system follows a closed-loop sensing-planning-acting pipeline. Multi-view visual observations and robot proprioception are used as policy inputs, teleoperated demonstrations provide expert supervision, and a fine-tuned SmolVLA policy generates bimanual action commands for autonomous skewering.

action trajectory, where each timestep contains camera observations, robot proprioceptive states, task information, and the corresponding expert action.

The teleoperation pipeline was designed to make data collection efficient and repeatable. Group members took turns operating the leader arms to collect demonstrations across different subtasks. Subtask labels were used to organize demonstrations by behavior type, such as picking up an object, adding an object to the skewer, or dropping an object into a cup. The physical leader-follower setup also made it easier for the operator to demonstrate smooth bimanual coordination than directly programming individual robot joint trajectories.

To make early training safer, cleaner, and more budget-friendly, foam balls were used as synthetic proxies for grapes during much of the data collection process. These proxy objects preserved the main geometric and manipulation requirements of the task while reducing the mess and variability of real fruit. The final dataset contains 272 episodes and 660,229 frames, totaling approximately 6 hours and 6 minutes of recorded demonstrations at 30 FPS. The visual observations include three RGB camera streams: a head camera, a left wrist camera, and a right wrist camera, each recorded at 640×480 resolution.

C. *Vision-Based Imitation Learning Policy*

The control policy is based on SmolVLA, a pretrained Vision-Language-Action model from the open-source LeRobot framework [1], [2]. We adapt this model to the fruit skewering task using imitation learning from the teleoperated demonstrations. At timestep t, the policy input is defined as

$$\mathbf{o}_t = \{\mathbf{I}_t, \mathbf{q}_t, \mathbf{l}\},$$

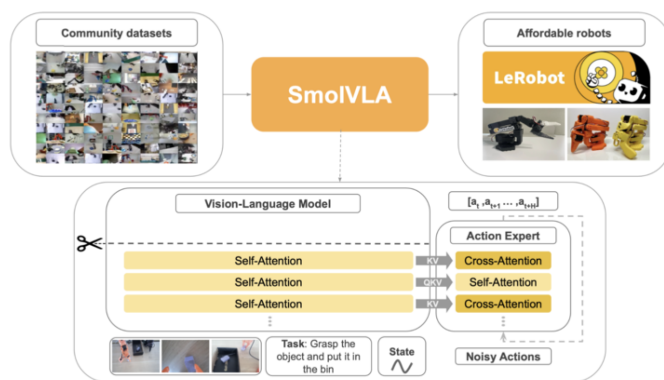


Fig. 3. SmolVLA-based imitation learning pipeline used in PIERCE, adapted from the LeRobot/SmolVLA framework [1], [2]. The pretrained Vision-Language-Action model is fine-tuned on teleoperated demonstrations and conditioned on visual observations, robot proprioception, and task-level language instructions to generate robot actions.

where $\mathbf{I}_t$ denotes the multi-view RGB observations, $\mathbf{q}_t$ denotes the bimanual robot proprioceptive state, and $\mathbf{l}$ denotes the natural language task instruction. This formulation allows the policy to jointly use visual scene information, robot state information, and task-level language conditioning.

Figure 3 illustrates the SmolVLA-based fine-tuning pipeline used in PIERCE. The pretrained policy provides a general visuo-motor prior, while our teleoperated demonstrations adapt it to the specific embodiment, camera viewpoints, and contact-rich skewering behaviors of our system.

The policy is trained to map each observation to continuous robot action commands. Example task instructions include “pick up grape,” “add held object to skewer,” and “drop held object in cup." Instead of manually specifying motion primitives for each subtask, the model learns the required bimanual coordination directly from expert demonstrations. This is important for skewering because the task involves precise timing, alignment, and contact interaction that are difficult to hand-code robustly.

We fine-tuned the SmolVLA policy using the standard training pipeline provided by LeRobot. Training was conducted on a 3×A100 GPU node for 8 epochs and reached convergence in approximately 18 hours. The fine-tuning process adapts the pretrained VLA model to our specific robot embodiment, camera setup, workspace, and culinary manipulation tasks.

D. *Real-Time Inference and Action Chunking*

During autonomous deployment, the policy must generate smooth action commands fast enough for physical robot execution. Instead of predicting only a single low-level action at each timestep, SmolVLA predicts a short-horizon action chunk,

$$\mathbf{A}_t = [\mathbf{a}_t, \mathbf{a}_{t+1}, \dots, \mathbf{a}_{t+k}],$$

where each $\mathbf{a}_t$ is a continuous bimanual robot action. This formulation is especially useful for skewering because the task requires temporally consistent motion during grasping, alignment, and insertion. Abrupt action changes can easily lead to missed contact, unstable alignment, fruit deformation, or object slipping.

At runtime, the robot executes the predicted action chunk while new camera observations and proprioceptive states are continuously collected. The policy is then queried again to produce an updated action chunk, forming a closed-loop perception-action pipeline. This allows the system to respond to small variations in object pose, gripper pose, and workspace configuration during autonomous execution. In our system, inference runs on a local GPU workstation at approximately 3 Hz, which is sufficient for the relatively slow and precise motions required by the skewering task.

Action chunking also helps reduce the practical effect of model inference latency. Since the policy predicts a sequence of near-future actions, the robot can continue executing smooth motion while the next inference step is being computed. When consecutive action chunks overlap, temporal ensembling is used to smooth the resulting commands and reduce abrupt transitions between policy outputs. This is important for bimanual manipulation because both arms must remain coordinated: the stabilizing arm must hold the skewer steady while the manipulation arm approaches, aligns, and inserts the object. Overall, action chunking makes the learned policy more suitable for real-time physical deployment than a purely step-by-step controller that waits for a new model output before every small action update.

E. *Runtime System and Dashboard*

To support real-time deployment and debugging, we implemented a runtime system built around a FastAPI server and a web dashboard. The server connects the trained policy, robot control interface, and camera streams. It receives current observations from the robot system, sends them to the policy for inference, and returns action commands for execution.

The web dashboard provides a live interface for monitoring and controlling the system. It displays the head camera and wrist camera streams, shows the current robot state, and provides task-level controls for commands such as picking up an object, adding an object to the skewer, and dropping an object into a cup. It also includes runtime controls for enabling or disabling robot actions, reconnecting the runtime system, and loading different trained checkpoints.

This dashboard was important for safe testing. Because the task involves physical contact between the grippers, food objects, cups, and the skewer, the operator needed a clear way to monitor policy behavior and stop execution when necessary. The model hot-swap function also made it easier to compare checkpoints and update the deployed policy without rebuilding the entire system. Overall, the runtime system connects the learned policy to the physical robot and makes autonomous execution practical in the real workspace.

V. Experiments and Results

A. *Evaluation Setup*

The system was evaluated in a tabletop workspace using the full PIERCE hardware setup, including the dual-arm robot, the 360° rotating base, the multi-view camera system, and the deployed SmolVLA visuo-motor policy. The primary objective of the evaluation was to measure whether the learned policy could perform the contact-rich bimanual behaviors required for autonomous fruit skewering, rather than only producing visually plausible arm motions.

We conducted three hierarchical stages of testing: (1) isolated grasping, (2) insertion precision, and (3) end-to-end skewer assembly. This staged structure allowed us to separate failures caused by grasping, alignment, insertion, and long-horizon task execution. Tests were first performed using synthetic foam proxies for safer and more repeatable validation, and final tests were performed using real grapes to evaluate how well the learned policy transferred to the target culinary manipulation setting.

B. *Isolated Skill Evaluation*

We first evaluated the robot's proficiency in the discrete sub-tasks required for skewering. These tests were designed to isolate the two most important low-level manipulation skills before evaluating the complete autonomous sequence.

- **Stage 1 (Single-Item Grasping):** The robot attempted to pick up individual objects from a bowl under randomized object positions. A trial was considered successful if the gripper secured the object, lifted it from the bowl, and maintained a stable grasp without visible slipping or crushing. We observed a **90% success rate** across repeated trials, suggesting that the visual policy could reliably localize and approach small objects in the workspace.

- **Stage 2 (Insertion Precision):** Starting from a successful grasp pose, the robot aligned the object with the skewer held by the stabilizing arm. A trial was considered successful if the object was axially aligned with the skewer and remained on the skewer after piercing. This stage achieved an **85% success rate**. Most failures occurred near the final insertion phase, where small lateral offsets or partial occlusions could prevent stable piercing.

These isolated evaluations showed that the policy learned the main manipulation primitives needed for the full task, while also revealing that insertion was more sensitive to visual alignment errors than grasping.

C. *End-to-End Autonomous Operation*

Stage 3 assessed the system's ability to assemble a complete 5-item fruit skewer autonomously. In this setting, the robot was required to repeatedly locate an item, pick it up, align it with the skewer, insert it, release the gripper, and prepare for the next item without human intervention. This evaluation was more difficult than the isolated stages because small errors could accumulate over the full sequence.

The system maintained an approximately **85% completion rate** for the full assembly. Successful trials demonstrated stable role coordination between the two arms: the stabilizing arm maintained the skewer pose, while the manipulation arm performed grasping, alignment, and insertion. This result indicates that the action-chunking policy was able to produce smooth bimanual motion over a longer horizon, rather than only succeeding in short isolated behaviors.

Compared with the isolated sub-task tests, end-to-end operation placed greater demand on robustness. The robot had to recover from small variations in object pose, gripper-object contact, and camera visibility across multiple consecutive actions. These tests therefore provided the strongest evidence that the learned policy could function as part of a complete autonomous manipulation pipeline.

D. *Synthetic Proxy Testing*

To ensure feasibility and hygiene, initial experiments were conducted using foam balls as synthetic proxies. These tests allowed us to benchmark the policy under controlled conditions with standardized geometry. The transition from these proxies to real fruit demonstrated the SmolVLA model's capability to handle moderate distribution shifts in material texture and weight with minimal additional data.

E. *Training and Runtime Performance*

The policy was fine-tuned offline on a 3×A100 GPU node for 8 epochs, totaling approximately 18 hours of training. The training process reached stable convergence, indicating that the model was able to learn the demonstrated bimanual action patterns from the teleoperation dataset. Since the demonstrations included grasping, alignment, skewering, dropping, and sorting behaviors, the trained policy learned a shared visuo-motor representation across multiple related manipulation skills.

For real-time deployment, the model ran on an NVIDIA 3090 GPU through the FastAPI runtime server. To mitigate VLA inference latency, we used temporal ensembling and action chunking, which maintained a control frequency of approximately 3 Hz. Although this frequency is lower than traditional low-level robot control loops, it was sufficient for the slow, precise, and contact-rich motions required by skewering.

In practice, the runtime system produced smooth and continuous motor execution. Action chunking allowed the robot to continue executing short-horizon motion while the next policy query was being computed. This was important for maintaining stable alignment during insertion, where pauses or abrupt command changes could easily lead to missed contact or object slipping.

F. *Failure Cases*

Through systematic testing, we identified two primary failure modes:

- During the final 2 cm of the insertion approach, the robot's own gripper occasionally obscured the target, leading to minor lateral misalignments.
- During end-to-end operation, residual fruit juice sometimes acted as an adhesive. In several trials, the robot successfully skewered the grape and opened the gripper, but the grape remained stuck to the interior of the claw. As the arm lifted to clear the workspace for the next item, the grape was inadvertently pulled off the skewer. This specific physical interaction was a primary cause of end-to-end failures, because it effectively nullified a previously successful insertion.

G. *Performance Summary*

The quantitative performance of the PIERCE system across the three hierarchical evaluation stages is summarized in Table II. Overall, the results show that the system performed reliably on isolated grasping and insertion, while the end-to-end task remained more challenging because errors could accumulate across multiple sequential actions. The results also indicate that the learned policy was able to transfer from proxy-object testing to real-fruit operation, although physical effects such as occlusion, slipping, and fruit adhesion still affected final task reliability.

TABLE II
QUANTITATIVE EVALUATION RESULTS FOR PIERCE AUTONOMOUS OPERATION.

Evaluation Stage	**Success Criteria**	**Success Rate**
1: Single-Item Grasping	Secure lift, no slipping or deforming	∼90%
2: Insertion Precision	Accurate axial alignment, stable piercing	∼85%
3: End-to-End Task	Autonomous assembly of 5-item skewer	∼85%

VI. DEMO VIDEO AND CODE REPOSITORY

A. *Demo Video Link*

The final demo video is available here: **PIERCE Final Demo Video**. The video demonstrates the project name, team members, data collection, testing, and autonomous skewering results.

B. Code Repository

The project code repository is available here: **PIERCE GitHub Repository**.

VII. Discussion and Future Work

A. Limitations

Despite the successful autonomous assembly of fruit skewers, the PIERCE system exhibited several key limitations rooted in the nature of imitation learning and the physical environment.

The SmolVLA policy, while effective within the training distribution, remained highly sensitive to distribution shifts. Even moderate variations in lighting or significant changes in the fruit's visual appearance could lead to poor generalization. Furthermore, the policy is currently purely Markovian, meaning it lacks a temporal memory of previous states. This absence of history context occasionally resulted in "looping" behaviors where the robot would repeat a failed maneuver without realizing the state remained unchanged.

Physically, the system was limited by the lack of tactile or force feedback. As observed in our results, the robot could not "feel" when a grape was stuck to the gripper due to residual juice leading to mechanical failures that a vision-only system struggled to perceive or correct.

B. Future Work

Future iterations of this project will focus on enhancing the robustness and cognitive depth of the system:

- **Reinforcement Learning (RL) Integration:** To move beyond the limitations of imitation learning, we propose using RL for fine-tuning. This would allow the robot to explore recovery behaviors and improve robustness in contact-rich phases through trial-and-error.
- **Memory and Context:** We plan to incorporate history context or textual memory into the VLA architecture. Adding a temporal window or a system-1/system-2 split would allow for higher-level agent orchestration and better long-horizon planning.
- **Synthetic Data Generation:** To address data scarcity, we aim to utilize planning algorithms to generate synthetic demonstration data. This would allow for training on a wider variety of edge cases without the time-intensive process of manual teleoperation.
- **Mechanical Optimization:** Utilizing the 360° rotating base more dynamically could allow the robot to re-orient the entire workspace to avoid visual occlusions. Additionally, exploring inductive biases or geometric priors in the model architecture could help the robot better understand the 3D spatial relationships required for precise skewering.

VIII. Statement of Contributions

Team members contributed to the design, implementation, and testing of the PIERCE system. The project required close integration between mechanical design, teleoperation, model training, runtime deployment, and experimental evaluation. The individual contributions are summarized in Table III.

TABLE III
Summary of individual contributions.

Team Member	Contributions
Qixin Xiao	Contributed to overall system design, visuo-motor policy formulation, SmolVLA fine-tuning, experimental analysis, hardware assembly, teleoperation data collection, physical testing, scene setup, PID tuning, dashboard/demo preparation, and report writing.
Avery Xi	Contributed to mechanical design, teleoperation data collection, physical testing, visuo-motor policy formulation, SmolVLA fine-tuning, inference pipeline development, system debugging, scene setup, PID tuning, dashboard/demo preparation, and final demo support.
Medhya Sundher	Contributed to data collection, subtask testing, hardware integration, runtime testing, and experimental evaluation. Helped validate isolated skills and end-to-end autonomous operation under both synthetic proxy-object and real-fruit settings, and report writing.
Elaina Mann	Contributed to hardware integration and mechanical design. Helped with final autonomous demonstrations and overall system preparation for the final demo.

In addition to these individual roles, all members participated in group discussions, and presentation preparation.

IX. Conclusion

This project presented PIERCE, a low-cost bimanual robotic system for autonomous fruit skewering. The system integrates custom mechanical design, multi-view visual perception, leader-follower teleoperation, imitation learning, real-time action chunking, and a runtime dashboard into a complete physical manipulation pipeline. By assigning complementary roles to the two arms, with one arm stabilizing the skewer and the other performing grasping, alignment, and insertion, PIERCE makes a difficult contact-rich culinary task more tractable for affordable hardware.

Our experiments show that a pretrained Vision-Language-Action policy can be adapted to a real bimanual robot through teleoperated demonstrations. The system achieved strong performance in isolated grasping and insertion tests and was able to complete end-to-end autonomous skewering in the final physical setup. These results demonstrate that modern visuo-motor imitation learning can support precise manipulation even without expensive tactile sensing.

At the same time, the project revealed important limitations. The policy remained sensitive to distribution shift, visual occlusion, and physical effects such as fruit slipping or sticking to the gripper. These challenges suggest that future versions of PIERCE should incorporate stronger temporal memory, better geometric priors, more diverse training data, and potentially reinforcement learning fine-tuning for recovery behaviors. Overall, PIERCE demonstrates a practical step toward affordable, vision-based, contact-rich culinary robotics.

REFERENCES

[1] Hugging Face LeRobot, "SmolVLA," https://github.com/huggingface/lerobot, 2024, accessed: Apr. 27, 2026.

[2] R. Cadene, S. Alibert, F. Capuano, M. Aractingi, A. Zouitine, P. Kooijmans *et al.*, "LeRobot: An open-source library for end-to-end robot learning," https://github.com/huggingface/lerobot, 2024, accessed: Apr. 27, 2026.

[3] J. Grannen, Y. Wu, S. Belkhale, and D. Sadigh, "Learning bimanual scooping policies for food acquisition," 2022, stanford University.

[4] J. Liu, Y. Chen, Z. Dong, S. Wang, S. Calinon, M. Li, and F. Chen, "Robot cooking with stir-fry: Bimanual non-prehensile manipulation of semi-fluid objects," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, 2022.

[5] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," 2023.

[6] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski *et al.*, "RT-1: Robotics transformer for real-world control at scale," 2022.

[7] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn *et al.*, "RT-2: Vision-language-action models transfer web knowledge to robotic control," 2023.

[8] M. J. Kim, K. Pertsch, S. Karamcheti *et al.*, "OpenVLA: An open-source vision-language-action model," 2024.